



BRAC UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science / CSE Program
Mid-Term Exam, Spring 2022

Course: CSE437 (Data Science: Coding With Real World Data), Section-1
Instructor: Dr. Md. Golam Rabiul Alam
Full Marks: 20
Time: 90 minutes

Note: There are Two questions, answer ALL of them. Course Outcome (CO), Cognitive Level and Mark of each question are mentioned at the right margin.

1. **Construct** a decision tree using the following bootstrap dataset for building a Random forest classifier to classify heart diseases. You have to consider all of the 4 features to construct the decision tree. Please show all the necessary calculations to build the decision tree. [CO2, C3, Marks:10]

Chest Pain	Good Blood Circ.	Blocked Arteries	Weight	Heart Disease
No	No	No	125	Yes
Yes	Yes	Yes	180	No
Yes	Yes	No	310	Yes
Yes	No	Yes	187	No

Table 1: Heart Diseases Dataset

2. **Find** the association rules using the Apriori algorithm considering the following transactions of a smartphone center. You have to show all the necessary computations to find out the association rules. You have to determine association rules of each and every frequent itemsets of length greater than or equal to 2. [Assume that, min_support=0.3334 and min_confidence=0.6, min_lift=1.2] [CO3, C4, Marks:10]

1	Smartpho	Charger	Earphone		
2	Smartpho	Charger			
3	Smartpho	Charger		Selfistick	Cover
4				Selfistick	Cover
5			Earphone		Cover
6	Smartphone			Selfistick	Cover